



Language model: movie reviewer

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Our task

This project aims to fine-tune a large language model to generate preliminary reviews and estimate the movie genre based only on the subtitles of a movie. This could reduce the number of badly perceived movies released each year caused by a bad story. We accomplish this by giving script reviewers a tool to reduce their workloads.

With the subtitle as the input, the model will return a JSON object:

- Rating prediction ranging from 1 to 10
- Genre or genres the movie belongs to.
- A review of the movie highlighting its strengths and weaknesses.

Dataset

Dataset Requirements

- Subtitles
- Reviews
- Genres
- Average ratings



Data Sources

- OpenSubtitles (base) - Extensive subtitle coverage with metadata.
- Amazon Movie Reviews - Diverse user reviews, keyed by movie title.
- IMDb Non-Commercial - Wider-distribution average ratings and official genres.

Cleaning Highlights

- Deduplication: Kept only the first subtitle per title to remove multiple releases.
- Subtitle filtering: Excluded partial/trailer-only subtitle files.
- Review mismatch: Noted misassigned reviews when movie and series share a title.

Final Dataset

- Movies: 32,979 unique titles
- Reviews: 1.87 M raw reviews → 98,937 kept (3 per movie)
- Split: 31,000 training / 1,000 validation / 1,000 testing

Preprocessing Workflow

- Formatted as JSON “prompt” (subtitles) + “completion” (rating, genres, one review)
- Trimmed middle of subtitle to fit model’s context - 31k tokens
- Wrapped in chat template and final tokenization for training

Solutions

Model: Gemma 3

- 4B parameters version
- Large context window: 32k
- Released: March 2025
- Exceptional generation fidelity

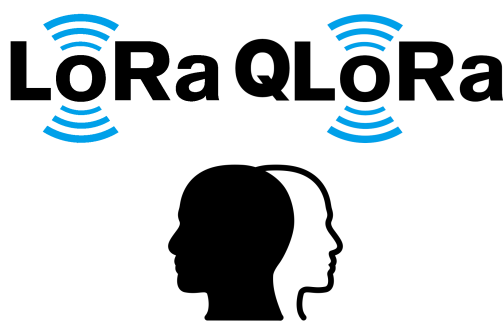


Model: Mistral

- 7B parameters instruct version
- Large context window: 32k tokens
- Released: May 22, 2024
- Advanced instruction-tuning

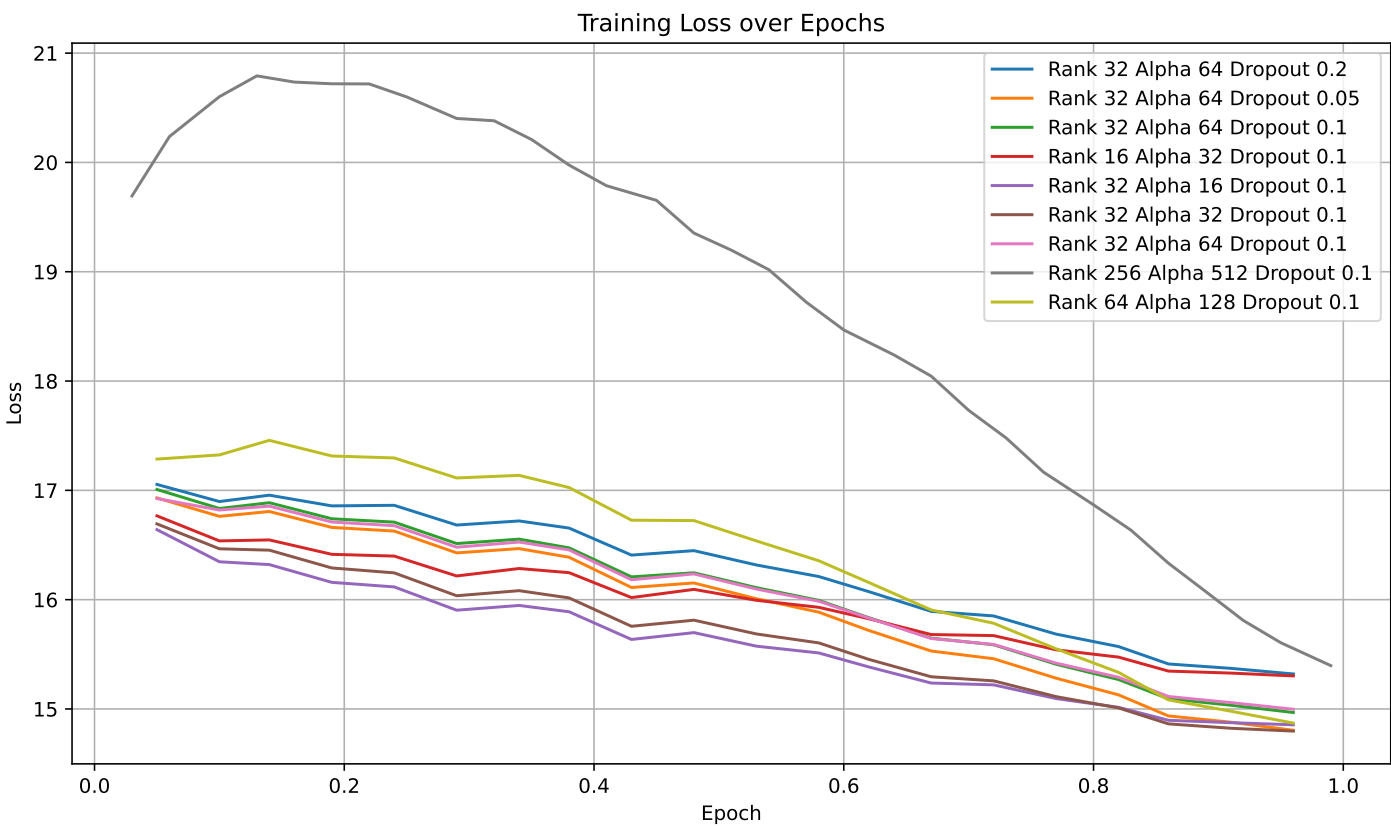


Finetuning



- 10 Gemma-3 models
- 1 Multi-Head Gemma-3 model
- 1 Mistral-7B model

Using various fine-tuning techniques



Tests

Score Prediction Accuracy

MAE, RMSE, % Within ± 1 of true score.

Genre Prediction Accuracy

Predicted genres against IMDb. Precision, Recall, F1-score.

Review Grammar Quality

Average number of grammar/style errors per review LanguageTool.

Review Similarity (to Human Reviews)

SBERT to get average cosine similarity score (0 to 1).

Sentiment Alignment

VADER sentiment analyzer, compare rating and sentiment.

Perplexity

Calculating the average perplexity of the models response.

Results

Model	MAE	RMSE	Sim.	Grammar	F1	Perp.
Deepseek	1.47	1.77	0.420	0.31	0.66	—
Base Gemma Instruct	2.46	2.76	0.284	0.36	0.30	19.53
Multi-Head	0.96	1.25	0.361	0.51	0.61	5.82
Gemma a32_r16	1.34	1.73	0.363	1.48	0.55	3.87
Gemma a8_r16	1.41	1.79	0.372	0.91	0.54	4.01
Gemma a32_r64 (0.05)	1.38	1.74	0.363	1.16	0.55	3.89
Gemma a512_r256	1.46	1.87	0.360	2.21	0.33	10.06
Mistral	1.90	2.34	0.350	0.23	0.40	12.88

Baselines vs. Top and Worst Models